

Who Is This Book For?

This book provides a friendly introduction to algorithms, with accompanying Python code. To get the greatest possible benefit from it, you should have some experience with the following:

Programming/coding. Every major example in the book is illustrated with Python code. We strive to provide walkthroughs and explanations of every code snippet to make the book digestible for someone with no Python experience and not much programming experience. Nevertheless, someone who has at least some basic understanding of the fundamentals of programming—such as variable assignment, `for` loops, `if/then` statements, and function calls—will be the most prepared to benefit.

High school math. Algorithms are often used to accomplish many of the same goals as math, like solving equations, optimizing, and calculating values. Algorithms also apply many of the same principles that are associated with mathematical thinking, like logic and the need for precise definitions. Some of our discussions veer into mathematical territory, including algebra, the Pythagorean theorem, π , and the teensiest bit of very basic calculus. We strive to avoid abstruseness and we don't venture beyond the math taught in American high schools.

Anyone who feels comfortable with these prerequisites should be able to master all the content in this book. It was written with the following groups in mind:

Students. This book is suitable for an introductory class on algorithms, computer science, or programming at the high school or undergraduate level.

Professionals. Several types of professionals could gain valuable skills from this book, including developers or engineers who want to gain familiarity with Python, and developers who want to learn more about the foundations of computer science and how to improve code by thinking algorithmically.